Lecture-27
Internal Flows VI: Flow Measurement Devices

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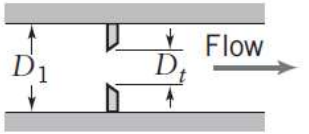
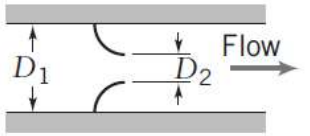
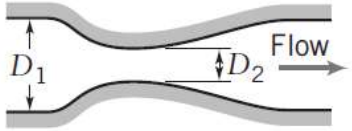
Learning Objectives

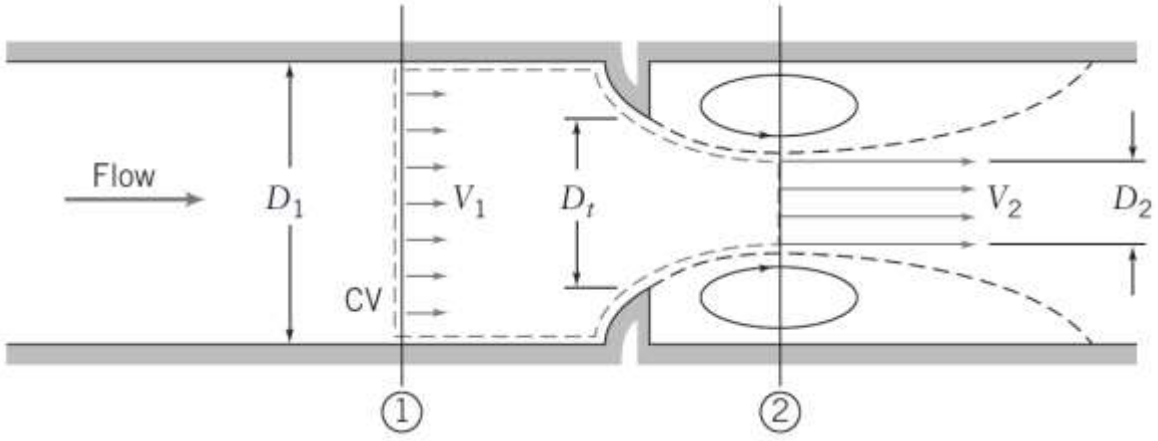
By the end of this lecture, you should be able to

1. Explain the **operating principles** of obstruction-type flow meters based on **Bernoulli and the continuity equation**.
2. Identify the differences in geometry between an **orifice plate**, a **flow nozzle**, and a **venturi meter**.
3. Define the **coefficient of discharge** and explain why it varies across different devices.

General operating principles

- The most common flow restriction devices are **orifices**, **nozzles**, and **venturi tubes**.
- Theoretically, all of them are analyzed using the **mass conservation** and **Bernoulli equations**.

Flow Meter Type	Diagram
Orifice	
Flow Nozzle	
Venturi	



$$\dot{m}_{\text{actual}} = \frac{CA_t}{\sqrt{1-(A_t/A_1)^2}} \sqrt{2\rho(p_1 - p_2)}$$

$$\dot{m}_{\text{actual}} = KA_t\sqrt{2\rho(p_1 - p_2)}$$

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

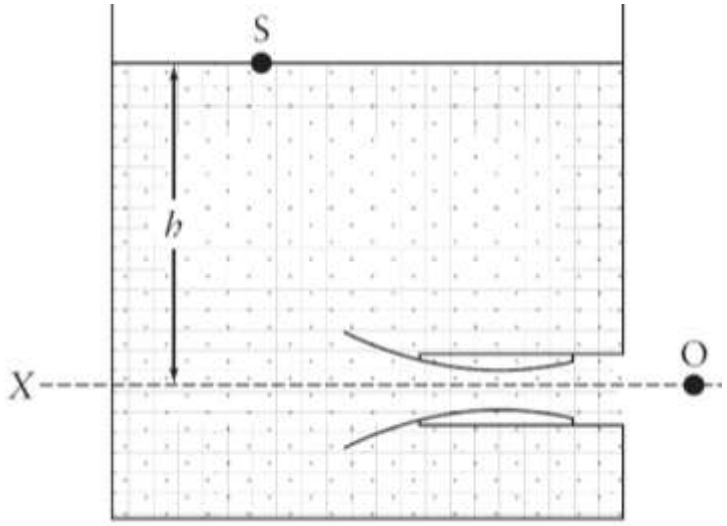
Flow through orifice

- The coefficient of velocity is defined as the **ratio of the actual velocity** of the jet through an orifice to the **theoretical velocity**.
- The coefficient of contraction is the **ratio of the area of the jet at the vena contracta** to the **area of the orifice**.
- The coefficient of discharge is the **ratio of the actual flow rate** to the **theoretical flow rate**.
- A **mouthpiece is a tube** that is fitted with a tank containing a liquid.

$$C_v = \frac{V}{\sqrt{2gH}}$$

$$C_c = \frac{a_c}{a}$$

$$C_d = C_v \times C_c$$



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Revisiting the learning objectives

1. Explain the **operating principles** of obstruction-type flow meters based on **Bernoulli and the continuity equation**.
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